

POLAKI LOKESWARA RAO

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in Artificial Intelligence and Machine Learning with a strong foundation in Python, Java, PostgreSQL, Machine Learning, and Generative AI. Experienced in developing end-to-end projects involving Machine Learning systems, LLM applications, Retrieval-Augmented Generation (RAG), AI chatbots, and intelligent recruitment systems. Familiar with LangChain, Qdrant, FastAPI, Flask, Streamlit, Hugging Face embeddings, and LLM APIs. Seeking an internship opportunity to apply AI, Machine Learning, and software development skills to real-world problems.

TECHNICAL SKILLS

Languages Backend: Python, Java, FastAPI, Flask

“ **GenAI / LLM Systems:** LLMs, RAG, LangChain, OpenAI APIs, Google Gemini APIs, OpenRouter, Prompt Engineering, Semantic Search, Data Retrieval, Reranking.

Machine Learning & Deep Learning: Machine Learning, Deep Learning, Feature Engineering, PyTorch, TensorFlow, Scikit-learn, NLP, OpenCV.

Databases & Vector Stores: PostgreSQL, Qdrant, Vector Embeddings

Data & MLOps: MLflow, Docker

Frontend & Tools: Streamlit, Git, GitHub, VS Code

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PROJECTS

Customer Churn Prediction & Retention Intelligence System

Python, Pandas, Scikit-learn, XGBoost, Random Forest, Flask, Streamlit, MLflow, Docker

- Built an end-to-end Machine Learning system to predict customer churn and identify high-risk customers for targeted retention campaigns.
- Performed exploratory data analysis, data validation, feature engineering, class-imbalance handling, cross-validation, and hyperparameter tuning across multiple classification models.
- Evaluated model performance using Precision, Recall, F1-score, ROC-AUC, PR-AUC, and confusion matrix, with threshold optimization for business-focused customer targeting.
- Developed a Flask-based inference service and Streamlit analytics dashboard for model predictions and interactive customer-risk analysis.
- Tracked Machine Learning experiments and model performance using MLflow to support reproducible model development and evaluation.
- Containerized the application using Docker and structured the project with modular training, inference, testing, and deployment components.

Converse – Intelligent AI Chatbot

Python, Flask, Streamlit, Google Gemini API, Ollama, Llama, DDGS, Git, Github

- Developed a conversational AI chatbot providing an interactive interface for asking questions, conducting conversations, and retrieving information.
- Built the backend using Python and Flask with REST-style endpoints for communication between the frontend and AI services.
- Integrated Google Gemini API for cloud-based AI responses and Ollama with Llama models for local AI inference.
- Implemented New Chat, Temporary Chat, Search Chats, and Recent Chats functionality with conversation-state management.
- Added session-based user authentication using Flask and implemented asynchronous frontend-backend communication using JavaScript fetch requests.
- Integrated DDGS web search to retrieve current information for relevant user queries and designed a responsive dark/glassmorphism-style chatbot interface.

AI Hiring Assistant

Python, Flask, RAG, LangChain, LLM, OpenRouter, Qdrant, Hugging Face, Streamlit, Vector Embeddings

- Developed an AI Hiring Assistant web application that helps recruiters upload, analyze, compare, rank, and interact with candidate resumes through a centralized hiring dashboard.
- Built an interactive hiring interface with features including Resume Upload, Job Description Upload, Chat with Resumes, ATS Resume Matcher, Resume Analyzer, Candidate Comparison, and Candidate Ranking.
- Developed a Flask-based Python backend to handle user authentication, dashboard routing, resume uploads, API requests, and communication between the application and AI services.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using LangChain, Qdrant Vector Database, and Hugging Face BGE embeddings to convert resume content into vector representations and retrieve relevant information.
- Integrated an LLM through OpenRouter to generate context-aware responses based only on retrieved resume information, enabling recruiters to query candidate skills, experience, and qualifications.
- Implemented resume analysis workflows for ATS matching, skill extraction, strengths and weaknesses identification, experience evaluation, candidate comparison, and ranking based on available resume information.
- Connected the complete workflow from resume upload to backend processing, vector database storage, RAG retrieval, LLM response generation, and chatbot interaction.

EDUCATION

Aditya Institute of Technology and Engineering College

Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence & Machine Learning) | CGPA: 7.7/10 Tekkali, A

D.R Junior College

Intermediate (M.P.C) | Percentage: 89%

2020 – 2022

Visakhapatnam, Andhra Pradesh

Keshava Reddy Schools

SSC | Percentage: 100%

2020

Srikakulam, Andhra Pradesh

CERTIFICATIONS

- Anthropic AI Fluency: Frameworks & Foundations – Anthropic
- Claude 101 – Anthropic
- Introduction to Machine Learning – Infosys Springboard
- Introduction to Deep Learning – Infosys Springboard
- Java Programming Certification
- Python Fundamentals Certification
- PostgreSQL: Became a SQL Developer – Certification of Completion

ACHIEVEMENTS

- Participated in AI and Machine Learning workshops to strengthen practical knowledge of Artificial Intelligence and Machine Learning concepts.
- Built multiple academic projects involving Machine Learning, Generative AI and intelligent AI applications.
- Continuously learning and exploring Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and AI Agents.